# Cloud Resource Sponsorship Request

## Open-Source ILT Platform for Early Screening of Learning and Literacy Disabilities

October 2026

**To:** The Research & Academic Grants Committee / Program Evaluation Team

Dear Program Directors and Evaluation Committee,

I am writing to respectfully request cloud infrastructure credits and hosting sponsorship for the **ILT (Invented Letter Task) Platform**—a global, non-commercial, open-source scientific initiative dedicated to the early identification and remediation of learning disabilities and literacy deficits among elementary school children worldwide.

## Scientific Background & Clinical Significance

Early identification of dyslexia, dysgraphia, and related literacy impairments is critical to preventing long-term educational marginalization. Grounded in peer-reviewed clinical research led by Dr. Hagit Hollander and Prof. Esther Adi-Japha (School of Education, Bar-Ilan University), our longitudinal studies of second-graders have proven that procedural motor skill learning (PLS) directly predicts end-of-year reading comprehension and written expression.

By analyzing fine-motor kinematics (sub-pixel path geometry, 4,096 levels of pressure, speed oscillation, and micro-tremor) as children acquire a novel graphic symbol ("Invented Letter"), the ILT paradigm assesses both:

1. **Initial Acquisition (Day 1):** The rate and quality of procedural motor learning across consecutive practice blocks.
2. **Consolidation & Automaticity (Day 14):** The offline retention and stabilization of the motor trace in long-term procedural memory without further training.

Crucially, our research demonstrates that this predictive model is particularly impactful for children from low socio-economic status (low-SES) backgrounds, who face the highest risk of undetected learning barriers.

## Mission & Global Humanitarian Impact

The ILT Platform is developed strictly as a **non-commercial, open-source public good**. Our mission is to democratize neuropsychological screening by transitioning from legacy, paper-based hardware to an accessible, distributed Web platform that can be deployed freely across public school systems, university research labs, and non-profit community clinics globally.

By eliminating proprietary licensing fees and hardware lock-in, the platform empowers educators and clinical diagnosticians to detect children at risk months before formal academic failure occurs, enabling timely, targeted interventions.

## Technical Architecture & Cloud Resource Requirements

The system has been architected from the ground up for high computational efficiency and strict data privacy (Privacy by Design):

- **Local-First P2P Field Operation:** During active screening in classrooms, tablets stream biometric stylus data to the evaluator's station via direct local Wi-Fi (WebRTC DataChannel), ensuring zero reliance on classroom internet bandwidth and sub-20ms latency.
- **Central Cloud Research Infrastructure:** We seek cloud hosting resources to support our containerized, open-source backend:
    - **Relational Core (PostgreSQL):** Managing administrative domains, research cohorts, and diagnostic session metadata.
    - **Time-Series Telemetry Engine (TimescaleDB):** Ingesting and querying millions of de-identified kinematic point samples, featuring native columnar compression that reduces storage footprint by up to 90%.
    - **Secure Object Storage (S3-Compatible):** Encrypted repository for observational video recordings and standardized block page templates.
    - **Central Analytics & Cohort Modeling:** Hosting open-source BI tools (e.g., Apache Superset / Metabase) to allow global researchers to identify cross-cultural patterns and train predictive machine learning models.

Because our core field pipeline operates locally and syncs asynchronously in batches, our compute and bandwidth footprint is lean, predictable, and highly cost-effective.

## Partnership & Mutual Value

In recognition of your sponsorship, our project will:

- Formally acknowledge your cloud platform's support in all forthcoming academic publications, clinical white papers, and conference presentations.
- Highlight your infrastructure sponsorship prominently within our open-source software repository (GitHub), project website, and documentation.
- Contribute benchmarked open-science datasets and case studies demonstrating social impact and healthcare accessibility on your cloud.

**Next step:** We would be grateful for the opportunity to submit a formal grant application through your academic or social-impact program, or to schedule a brief introductory briefing to present the platform.

Thank you very much for your time, consideration, and ongoing support of scientific innovation and educational equity.

Respectfully and sincerely yours,

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